

# IBD T-Cell Marker Analysis

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# Dataset

- **Target:** inflammatory bowel disease (Crohn's disease + ulcerative colitis)
- **BG:** when the immune system attack healthy tissue -> diarrhea & nausea
- **Condition:** CD / UC / Control (CK)
- **Sample:** 12 samples, 4 for each condition

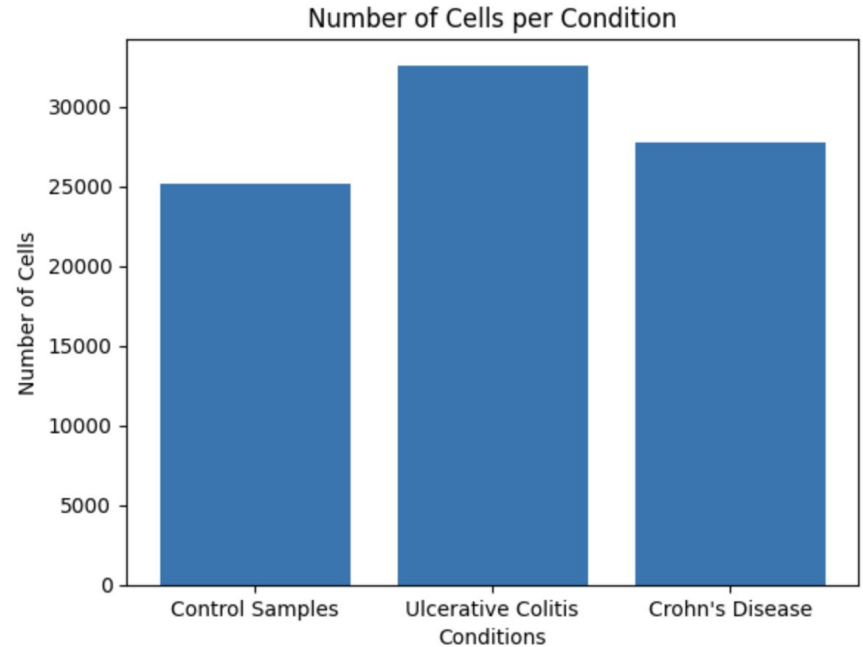
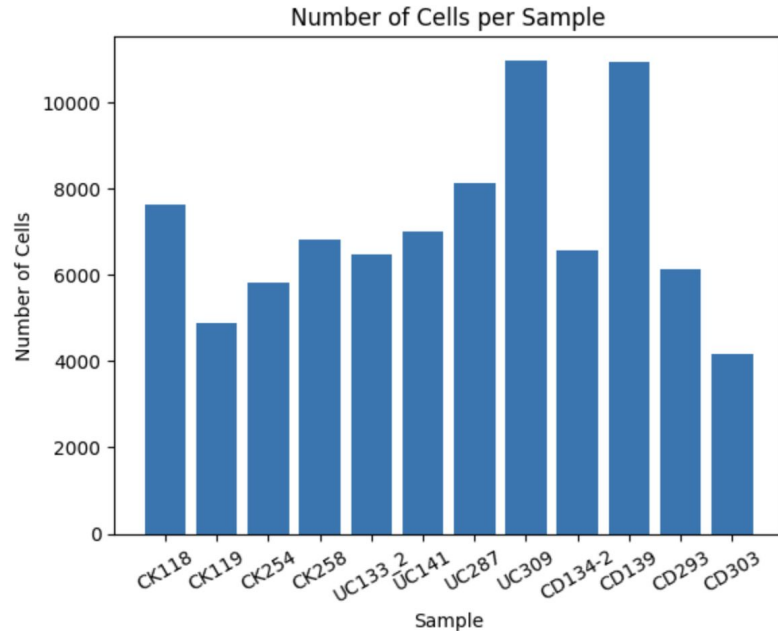
|       | Condition       | Sample | Cell_Index | CD103 ITGAE AHS0001 pAbO | CD127 IL7R AHS0028 pAbO | CD161:DX12 KLRB1 AHS0002 pAbO | CD195:2D7 CCF |
|-------|-----------------|--------|------------|--------------------------|-------------------------|-------------------------------|---------------|
| 0     | Control_samples | CK118  | 707743     | 12                       | 23                      | 4                             |               |
| 1     | Control_samples | CK118  | 577396     | 12                       | 21                      | 11                            |               |
| 2     | Control_samples | CK118  | 89979      | 17                       | 18                      | 6                             |               |
| 3     | Control_samples | CK118  | 57379      | 13                       | 11                      | 5                             |               |
| 4     | Control_samples | CK118  | 658439     | 6                        | 17                      | 4                             |               |
| ...   | ...             | ...    | ...        | ...                      | ...                     | ...                           | ...           |
| 27774 | Chrons_Disease  | CD303  | 72516      | 4                        | 28                      | 2                             |               |
| 27775 | Chrons_Disease  | CD303  | 526362     | 9                        | 2                       | 30                            |               |

# Project Workflow

- **Question:** Do intestinal T cells from Crohn's disease, ulcerative colitis, and control samples differ in selected marker patterns, especially in the proportion of cells with a CDpop-like phenotype?
- **Step 1:** Preprocessing
  - Load/ add label columns - - check N/A - - merge
- **Step 2:** Summary by sample and condition
  - Bar plots to show the shape
- **Step 3:** Comparison
  - Compare selected protein/RNA markers across conditions
  - Differential-expression analyses and volcano plots + heat map.
- **Step 4:** Interpretation and limitations

## Figures 1 & 2: Data summary

- Not an equal representation of all cells
- Samples have different amount of cells



## Workflow

- Load in Data
- Sum Total Gene Expression for Each Cell
- Gene A expression/Total Cell Expression
- \* 1,000,000 → Counts Per Million matrix
- Log transformation: `log2_counts = np.log2(cpm_matrix + 1)`

```
df_control = log2_counts[group_labels == 'Control_samples']  
df_CD = log2_counts[group_labels == 'Chrons_Disease']  
df_UC = log2_counts[group_labels == 'Ulcerative_colitis']
```

- Grouped my condition
- Calculated means of each gene by group
- Subtract control mean from disease means
  - yields if groups are more/less expressed relative to control
  - CD mean - control mean
  - UC mean - control mean
- Up reg: expression > 1; down regulated: expression < 1

The diagram illustrates the mathematical identity  $\log\left(\frac{x}{y}\right) = \log(x) - \log(y)$ . It features three curved arrows: a green arrow from  $x$  to  $\log(x)$ , an orange arrow from  $y$  to  $\log(y)$ , and a blue arrow from the fraction  $\frac{x}{y}$  to  $\log\left(\frac{x}{y}\right)$ .

$$\log\left(\frac{x}{y}\right) = \log(x) - \log(y)$$

Figure 3. Significant Gene Expression Changes Relative to Control Baseline

### Thresholds for Genes:

- Log4 fold change:  $\log_2(4.0)=2$
- P value: 0.0001

### Interpretation:

- Unique gene expression
  - Not many “shared” up/down regulated genes
- Ulcerative Colitis: more extreme down regulated gene expression compared to control
- Crohn's Disease: fewer Up/Down regulated genes

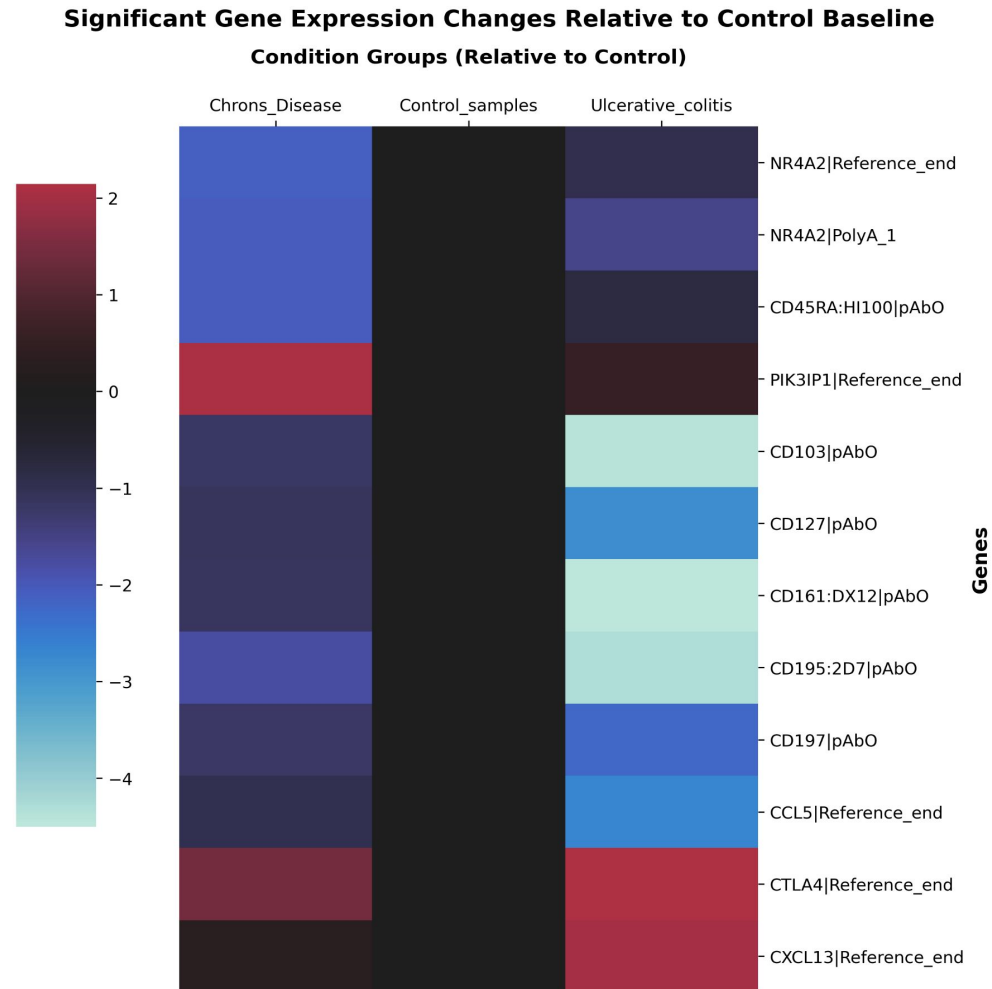
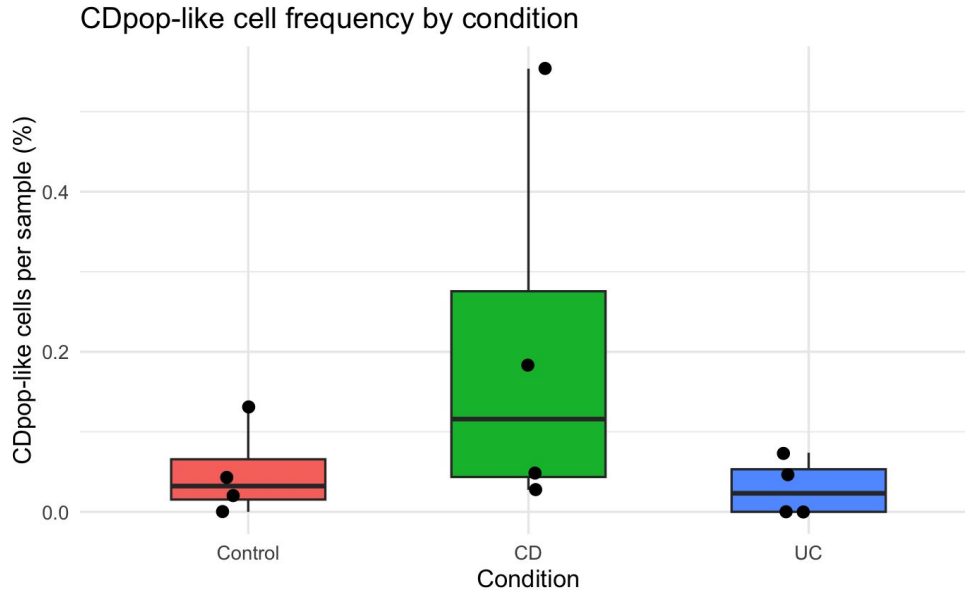


Figure 4: CDpop-like cell frequency by condition



### CDpop-like cell definition:

- $CD4^+ CD103^+ CD161^+ CCR5^+ CD27^- CCR7^-$
- Each point represents one sample

### Interpretation:

- CD samples showed a higher CDpop-like frequency than Control and UC
- Differences were not statistically significant
  - Pairwise Wilcoxon tests with BH correction: all adjusted p-values > 0.05

# Conclusion, Limitations and Future work

## Conclusions:

- Observed different genes expressed in CD and UC samples compared to Control samples (figure 3)
- Observed a higher frequency of CDpop-like cells in CD samples compared to Control and UC samples (figure 4)

## Limitations:

- Four samples per condition (for sample-level comparison)
- Simplified detected/ not-detected thresholds
- P-value interpretation

## Future work:

- Investigate pathways correlated with genes
- Apply biologically validated thresholds



THANK YOU!